

Unit Interval Parking Functions

Math 724, Fall 2025

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As in [3], we begin with parking functions. Suppose we have n cars in a line, ready to park on a one-way, one lane street with n parking spaces numbered $1, 2, \dots, n$. Each of these cars has a favorite parking spot, which we call their *preference*, and the i -th car's spot appears in the i -th spot of the *preference vector*. Each car enters the street, and drives to its preference. If that spot is open, the car parks there. If not, the car parks in the next available spot (if one exists). If a car's preference and all following spots are taken, the car cannot park. However, if all cars can park, we call this preference vector a *parking function*.

Let a_i denote car i 's preference. Then, $\mathbf{a} = (a_1, a_2, \dots, a_n) \in [n]^n$ is a preference vector. If all cars park, then we call $\mathbf{a}$ a parking function.

We generalize to interval parking functions [1]. Here, we construct another vector $\mathbf{b} = (b_1, b_2, \dots, b_n)$. We say that $(\mathbf{a}, \mathbf{b})$ is an *interval parking function* if every car parks in spot p_i where $a_i \leq p_i \leq b_i$.

Question 1 Confirm $((1, 3, 2), (2, 3, 2))$ and $((3, 1, 3, 1), (4, 4, 4, 2))$ are interval parking functions.

Question 2 Confirm that $((1, 3, 2), (2, 2, 2))$ and $((1, 3, 3), (3, 3, 3))$ are not interval parking functions.

Question 3 What about $((1, 4, 1, 5, 1, 4), (2, 5, 2, 6, 3, 5))$ and $((5, 6, 1, 1, 2, 3), (6, 6, 2, 2, 3, 4))$?

Question 4 Think about $\mathbf{b}$. In a way, b_i gives an upper bound for the parking spot of car i . How might we specify b_i for all $i \in [n]$ to recover classical parking functions from interval parking functions?

Question 5 Consider now the unit interval $[0, 1] \subseteq \mathbb{R}$; it has length 1. How might we adapt $\mathbf{b}$ to define a unit interval of preferences for each car i ? Use this to define a **unit interval parking function**. (I defined these in my undergraduate thesis [2]. Call me over to see if we have the same definition!)

Question 6 Confirm $((1, 3, 2), (2, 3, 3))$ and $((5, 6, 1, 1, 2, 3), (6, 6, 2, 2, 3, 4))$ are unit interval parking functions. Confirm $((3, 1, 3, 1), (4, 2, 4, 1))$ is not. What about $((5, 6, 1, 1, 2, 3), (6, 6, 2, 2, 3, 4))$?

Question 7 List all unit interval parking functions for $n = 1$, $n = 2$, and $n = 3$. Create a table to record n and $|\{\text{unit interval parking functions}\}|$.

Question 8 How can we collect more data to find $|\{\text{unit interval parking functions}\}|$ when $n \in \{4, 5, 6\}$? Maybe write your own pseudo-code or real code? We can also discuss together!

Question 9 Go to <https://oeis.org/>.

1. Enter your sequence into the Online Encyclopedia of Integer Sequences!
2. Write the OEIS code for your sequence.
3. List (and explain) three things that this sequence counts.
4. Read (and be able to explain) the abstract for one paper cited in this OEIS entry.

Question 10 Choose an (interesting) object that the sequence enumerates. Construct a bijection between the set of unit interval parking functions and the set of objects that you chose.

References

- [1] E. Colaric, R. DeMuse, J. Martin, and M. Yin. Interval parking functions. *ArXiv*, 2020. <https://arxiv.org/pdf/2006.09321.pdf>.
- [2] K. P. Hadaway. *On Combinatorial Problems Related to Generalized Parking Functions*. Bachelor's Thesis, Williams College, 2021.
- [3] A. Konheim and B. Weiss. An occupancy discipline and applications. *SIAM Journal on Applied Mathematics*, 14(6):1266–1274, 1966.